

Pav Bhaji Classification From Text

Detecting a single dish among 1,500 social-media food posts using caption and hashtag metadata

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1,500	18	12.2%
POSTS ANALYSED	RAW METADATA FIELDS	MINORITY CLASS

Summary

This project tests a practical question: **can a dish be identified from what people write about a photo, without ever looking at the photo?** Working with 1,500 scraped social-media food posts, I built an end-to-end text-analytics pipeline parsing nested JSON, engineering a combined caption-plus-hashtag feature, cleaning multilingual and emoji-heavy text, and vectorising it with TF-IDF then benchmarked three classifiers against a heavily skewed target.

The headline finding is a cautionary one, and it is the most valuable result in the report: **the two baseline models reached 88% accuracy while identifying zero examples of the target class.** Accuracy was being inflated entirely by the majority class. Only a cost-sensitive gradient-boosting model produced any real signal on the minority class, and diagnosis of the data itself not further model tuning turned out to be the genuine bottleneck.

Objective & Business Context

Food-delivery platforms, restaurant aggregators and social listening teams all need to tag large volumes of user-generated food content at low cost. Image models are accurate but expensive; captions and hashtags are already available as plain text. The question this analysis answers is whether that cheap metadata alone is discriminative enough to be useful.

- **Target:** Binary label - Pav Bhaji (1) vs Non-Pav Bhaji (0).
- **Predictors:** Post caption plus the full hashtag string, combined into a single text field.
- **Success criterion:** recall and F1 on the minority class deliberately not overall accuracy.

Dataset Overview

The raw data arrived as a nested JSON export of social-media posts alongside two folders of downloaded images, which supplied the ground-truth labels.

Attribute	Value	Analytical note
Post records	1,500	18 fields, several deeply nested
Images available	452	Labels derived from folder structure
Pav Bhaji (class 1)	183	12.2% of records — severe skew
Non-Pav Bhaji (class 0)	1,317	87.8% — dominates every metric
Class ratio	7.2 : 1	Drives the model bias seen later
Text features after TF-IDF	5,000	Sparse, high-dimensional matrix

Two structural issues surfaced immediately and shaped the whole analysis: the label source is thinner than the record set — only 452 images exist for 1,500 posts, so any record without a matching image defaults to class 0 and introduces label noise and the captions are multilingual and emoji-dense, mixing English, Hindi and Devanagari script within single posts.

Data Preparation Pipeline

Each stage below was designed to move the data from unusable nested objects to a clean numeric matrix, and each one surfaced a decision worth recording.

Stage	What was done	Outcome
Ingestion	Unpacked the archive; loaded nested JSON into a DataFrame	1,500 × 18 base table
Extraction	Pulled caption text out of nested edge objects; flattened hashtag lists to strings	Two clean text columns
Key building	Parsed image filenames out of media URLs with regex; kept post shortcode as a stable ID	Reliable join key
Labelling	Mapped images from their source folders and joined labels back to post records	Ground-truth target column
Cleaning	Lower-cased, stripped punctuation and symbols, tokenised, removed English stop words	Normalised text field
Vectorisation	TF-IDF with a 5,000-term vocabulary cap	Model-ready feature matrix

Feature Selection: Why Only Caption, Tags & ID

The raw JSON exposed 18 fields per post, but only three were carried into the modelling table:

caption_text, **tags_text**, and **image_id**. This was a deliberate scoping decision, not an oversight.

The objective was **text-based classification** — identifying a dish from what people wrote about a photo, without looking at the photo itself. That framing sets a clear inclusion test: a field earns a place in the feature set only if it is a human-readable description of image content. Caption and hashtags pass that test directly; nothing else in the schema does.

We connected image files with their corresponding JSON metadata by extracting filenames from the image paths and the `display_url` within the JSON. This allowed us to create a unified DataFrame where each image's metadata was associated with its correct label (Pav Bhaji or Non-Pav Bhaji), derived from the directory where the image was stored. This process was crucial for establishing the ground truth for our classification task.

Dropped field(s)	What it captures	Why it was excluded
<code>dimensions</code> , <code>is_video</code> , <code>video_view_count</code>	Numeric / boolean media metadata	Not textual content — describes the file, not the dish
<code>display_url</code> , <code>thumbnail_src</code>	Links to image resources	Structural pointers, not descriptions
<code>edge_liked_by</code> , <code>edge_media_to_comment</code>	Engagement counts	Reflects popularity, not subject matter
<code>location</code> , <code>owner</code>	Where / who posted	Context about the post, not what is depicted
<code>taken_at_timestamp</code>	Upload time	Temporal metadata, unrelated to dish identity

`image_id` (the post shortcode) was kept, but not as a predictive feature it is a join key. Its only role is linking a processed text record back to its source image so the correct label survives the merge; the model never sees it as an input.

Narrowing to caption and tags rather than engineering every available field into the model was itself a design choice with three payoffs: it keeps the feature space tied to a single, defensible hypothesis (text describes content); it avoids injecting noise from fields with no plausible causal link to the label, which would only dilute the TF-IDF signal further given how sparse the minority class already is; and it keeps the resulting model interpretable any keyword the model weights heavily can be traced straight back to something a human actually wrote. Fields like engagement or location may be valuable for other questions, such as predicting popularity, but they were out of scope for this one.

Key Analytical Findings

1. The class imbalance is the dominant signal in the data

At a 7.2 : 1 ratio, a model that predicts "not Pav Bhaji" every single time scores 87.8% accuracy without learning anything at all. That number is almost exactly what the baselines produced a textbook accuracy paradox, and the reason the evaluation was reframed around per-class recall.

2. The text itself is weakly discriminative

Pav Bhaji hashtags travel in crowds. The same posts carry generic food tags street food, Indian food, foodie, Mumbai that appear just as often on the negative class. Because TF-IDF down-weights terms common across the corpus, the handful of genuinely distinguishing tokens is diluted in a 5,000-dimension sparse space where the minority class occupies very little of the volume.

3. Label quality is the real ceiling

Roughly two-thirds of the post records have no matching image file and were therefore assigned the negative label by default. A meaningful share of those are almost certainly Pav Bhaji posts sitting in the wrong class. **No amount of resampling or model tuning can recover from labels that are wrong at the source** which reframes this from a modelling problem into a data-collection problem.

Modelling Approach

Data was split 80/20 with stratification to preserve class proportions (1,200 training / 300 test records, of which 37 test records are positive). Three approaches were benchmarked.

- **Logistic Regression** fast, interpretable linear baseline on the raw imbalanced data.
- **Random Forest with synthetic oversampling** SMOTE, BorderlineSMOTE and ADASYN were each trialled inside a leak-safe pipeline, with neighbourhood sizes and sampling ratios tuned, plus balanced class weights.
- **XGBoost with cost-sensitive learning** instead of fabricating synthetic minority rows in sparse text space, the positive class was re-weighted at 7.22 directly in the loss function.

Oversampling was deliberately confined to the training fold inside the pipeline, so no synthetic information leaked into evaluation.

Results

Model	Accuracy	Precision (1)	Recall (1)	F1 (1)
Logistic Regression	0.877	0.00	0.00	0.00
Random Forest + SMOTE family	~0.88	0.00	0.00	0.00
XGBoost (cost-sensitive)	0.843	0.25	0.14	0.18

The best model is the one with the lowest accuracy. Logistic Regression and every Random Forest variant collapsed onto the majority class and detected none of the 37 positive test cases. XGBoost gave up roughly three accuracy points and, in exchange, became the only model to make correct positive predictions at all. Judged on the metric that matches the business question, it is the clear winner.

Notably, synthetic oversampling never moved the needle here. Interpolating between minority points in a 5,000-dimension sparse matrix produces samples that sit in near-empty space and carry little usable signal cost-sensitive weighting proved the better tool for this data shape.

Limitations & Recommendations

Ranked by expected return on effort:

- **Fix the labels first.** Restrict the modelling set to the 452 records with verified image labels, or source images for the remaining posts. This is the single highest-value action and no modelling change competes with it.
- **Replace TF-IDF with semantic embeddings.** Sentence-transformer or multilingual BERT representations would capture meaning across the English–Hindi mix that bag-of-words treats as unrelated tokens.
- **Tune the decision threshold, not just the model.** Optimising the probability cut-off against a precision–recall curve typically recovers substantial minority recall at no training cost.
- **Report PR-AUC and F1 as the headline metrics.** Accuracy is actively misleading at this class ratio and should not appear in a stakeholder summary.
- **Combine signals.** Pairing text features with a lightweight image model would likely outperform either source alone.

Skills Demonstrated

Area	Applied in this project
Data wrangling	Nested JSON flattening, regex key extraction, multi-source joins, deduplication
Text analytics	Multilingual cleaning, tokenisation, stop-word removal, TF-IDF vectorisation
Machine learning	Stratified splitting, leak-safe pipelines, SMOTE / BorderlineSMOTE / ADASYN, cost-sensitive boosting
Evaluation	Per-class precision, recall and F1; diagnosing the accuracy paradox
Analytical judgement	Identifying label quality — not algorithm choice as the true constraint

Conclusion

Caption and hashtag metadata alone are not yet sufficient to isolate a single dish within a corpus of visually and linguistically similar food posts. The pipeline, however, is sound and ready to run again the moment the labelling gap is closed.
